HuBisCO

Exploring alternative substrate compatibility in RuBisCO-mediated carbon fixation

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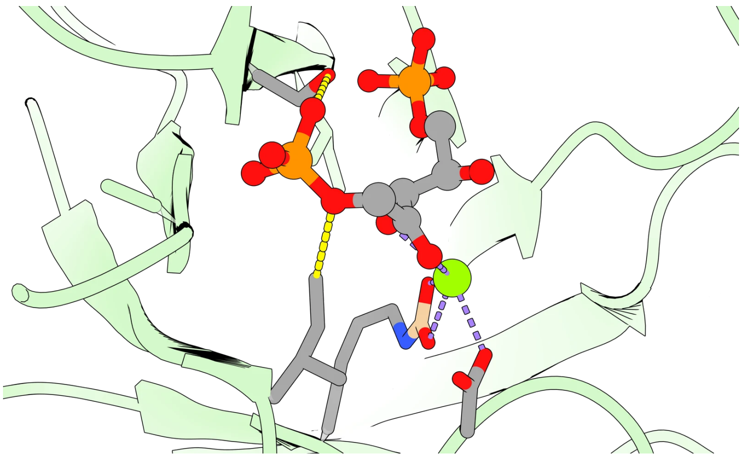
THE RESULT

Alternative-substrate turnover was not established. The study produced experimental tools, identified active-site hypotheses and exposed substrate-access and assay limitations that define the next experiments.

01 The question

Could a six-carbon sugar-phosphate substrate help redirect RuBisCO chemistry? HuBP was the intended target of a proposed alternative carbon-fixation route. The project examined whether the active site of Rhodospirillum rubrum Form II RuBisCO could support this chemistry.

02 From structure to testable substitutions



I164T
Introduces a hydroxyl group and changes local steric bulk.

S368A / S368C
Alter the side-chain chemistry near the phosphate-binding environment.

Project-supplied active-site rendering. 9RUB is a deposited wild-type reference with native RuBP; it is not a solved HuBisCO mutant structure.

03 Experimental route

Structural comparison and docking informed candidate substitutions. Recombinant WT, I164T, S368A, S368C and PHI protein work supported NADH-linked coupled assays and phosphorus-31 NMR analysis of precursor chemistry.

What the evidence supports

04 Three complementary readouts

Protein work

WT and three RuBisCO variants, plus PHI, were expressed and purified. Native RuBP-dependent activity provided an operational assay reference.

Coupled assays

HuBP was not experimentally accessible. Exploratory Hu6P / Ru5P conditions did not establish meaningful alternative-substrate activity. Weak mutant signals were inconclusive.

Precursor chemistry

Phosphorus-31 NMR tracked PHI-mediated F6P / Hu6P interconversion. The peak assigned to Hu6P increased relative to F6P; this is not evidence of RuBisCO turnover.

REPORTED NMR PEAK-AREA RATIOS

24 h	**0.035**	144 h	**0.196**

Hu6P:F6P assignments reported in the dissertation. Two reported time points do not establish a kinetic mechanism or a mechanistic rate constant.

05 Limitations that determine the next experiment

Substrate access: obtain and verify HuBP before directly testing the central hypothesis.
Attribution: repeat matched WT / mutant comparisons with no-enzyme and supporting-enzyme controls; the exploratory mutant assays lacked replication.
Normalization: distinguish total protein from active enzyme.
Confirmation: collect a denser NMR time course and independently confirm product identity.

06 My contribution and research practice

Recombinant expression and purification; coupled-assay development and analysis; structure-guided mutant evaluation and docking. The project integrated structural reasoning with experimental evidence to prioritize follow-up work. NMR support is acknowledged in the dissertation.

Sources: Bill Huang, project dissertation, Is RuBisCO “HuBisCO”?, Summary, Results and Methods; supplied research CV. Experimental reference structures: RCSB PDB 9RUB and 5RUB. Original dissertation retained privately.

betoadfish.github.io